

Reality Probe: Information-Limited Physical Verification of Captured Worlds

SWAYAM SINGAL

A captured scene can now carry enough physical state to be simulated, edited, and replayed after capture. That makes physical editing possible, but it also creates a verification problem: a rewrite can reduce ordinary held-out error while making an untouched physical behavior worse. We study this failure mode and introduce Reality Probe, a rewrite-level verification contract that separates proposal generation from a reserved physical test and returns SUPPORT, VETO, or UNRESOLVED according to a fixed evidence rule. The experiments deliberately include both successful and failed verification channels. Dark-Field Counterfactual Spectroscopy (DCS), a signed intervention contrast used inside Reality Probe, passes a constructed control but resolves none of the proposals in several frozen synthetic regimes. A separately frozen known-force compliance channel increases median standardized evidence magnitude by $11.46\times$, yet all 36 fresh synthetic proposals remain below the decision boundary. On a repository-locked IRIS pendulum final set, a finite-amplitude period probe makes 90/110 controlled decisions correctly, compared with 40/110 for a matched small-angle diagnostic baseline; the 110 cases are nested within 10 physical videos, and all 10 placebo cases are correctly left UNRESOLVED. The result does not transfer automatically to another equation family. A frozen IRIS free-fall validation run passes low-level quality checks on 5/5 videos but reaches 111.7% median acceleration relative error, so the lane is closed and its final split remains unopened. These results support a conditional systems claim: verification is useful when the reserved physical channel contains information that separates the candidate rewrite from the baseline, and abstention is the correct outcome when it does not.

Additional Key Words and Phrases: captured worlds, inverse physics, physical verification, counterfactual simulation, material point method, uncertainty, experiment design, Gaussian splatting

1 INTRODUCTION

Consider a captured object whose simulated material parameters have been fitted from video. An optimizer proposes a new parameter set because it lowers error on ordinary held-out frames. That improvement answers a fitting question: the rewrite reproduces more of the observed motion. It does not yet answer the physical question that matters before commit: will the edited world behave more correctly under a different load, trajectory, or interaction?

That distinction has become practical because scene representations increasingly carry executable physical state. Physics-aware neural fields and Gaussian representations now support dynamic reconstruction, material inference, and interaction synthesis [Li et al. 2023; Liu et al. 2025; Rho et al. 2026; Wada et al. 2026; Xie et al. 2024; Zhang et al. 2024]. Once a scene can be edited physically, its update path needs an evidence rule for deciding whether a proposed rewrite should be accepted.

The difficulty is informational. Several physical models can reproduce similar visible motion while disagreeing under another experiment. Parameter compensation, constitutive assumptions, transfer schemes, boundary conditions, and numerical choices can all contribute to that ambiguity. Structural identifiability studies whether the available observations determine the model parameters; optimal

experiment design asks which measurements or interventions are informative; verification, validation, and uncertainty quantification study whether a model is credible for its intended use [Asadi and Laksari 2025; Laudy 2026; Oberkamp et al. 2007; Wang et al. 2026b]. Reality Probe builds on those ideas at a narrower operational level: a persistent captured world already has a candidate rewrite, and the system must decide what the reserved physical evidence says about that specific change.

We call a rewrite *deceptive* when its ordinary held-out loss improves while an untouched physical target becomes worse. The word describes the evidence pattern, not the intent of the optimizer. Figure 1 summarizes the transaction. A proposal is produced from fitting evidence, a different physical question is asked after the proposal is fixed, and the resulting evidence is standardized against its uncertainty. One frozen synthetic case makes the failure mode concrete: ordinary held-out error decreases by 14.93%, while the untouched-target error increases by 9.03%. That case was selected after the experiment for visualization and does not enter the aggregate statistics or thresholds.

Reality Probe turns approval of such a rewrite into an explicit evidence transaction. Proposal generation uses fitting data and ordinary held-out observations. Verification begins after the proposal is fixed and uses a separately reserved physical channel. The verifier compares the baseline and candidate against that evidence, carries measurement, repeat, and numerical uncertainty, and returns SUPPORT, VETO, or UNRESOLVED. SUPPORT indicates that the candidate improves the reserved physical error by at least the frozen evidence margin; VETO indicates the opposite; UNRESOLVED records that the available experiment does not separate the two strongly enough.

Dark-Field Counterfactual Spectroscopy (DCS) is one probe tested inside this contract. DCS forms signed combinations of intervention responses so that lower-order common behavior cancels and higher-order disagreement becomes easier to observe. The cancellation machinery comes from classical finite-difference and inclusion-exclusion constructions. The scientific question here is empirical: does that type of contrast provide enough information to verify a captured-world rewrite under frozen uncertainty and decision rules?

The answer is mixed, and the experimental program is designed to preserve that fact. DCS passes a constructed positive control, then fails to produce resolved evidence in frozen truth-world validation, witness-space refinement, and a pair-specific repair-veto population. Matched raw, Fisher-style, and maximum-motion comparators also remain unresolved in the repair-veto stage. A new known-force compliance channel is then frozen before execution. It raises median standardized evidence magnitude by $11.46\times$ on fresh synthetic truth worlds but still leaves all 36 proposals below the decision magnitude.

Measured video provides both a positive result and a boundary case. On 10 fresh IRIS pendulum_90 videos, the finite-amplitude



Fig. 1. Reality Probe at a glance, composed from frozen result panels. A candidate repair can improve ordinary held-out fit while worsening an untouched physical target. The repair is then asked a different physical question, its motion response is compared with the reference response, mechanism-level residuals are exposed, and the evidence is standardized. The displayed case is post hoc for visualization; the 11.46 $\times$ signal gain and 0/36 resolved coverage are aggregate fresh synthetic force-compliance results.

period probe makes 90/110 controlled decisions correctly, compared with 40/110 for the small-angle diagnostic baseline, with all 10 placebo cases left UNRESOLVED. The 110 rows are nested controlled cases across 10 videos; they are not 110 independent physical experiments. A separately frozen IRIS free-fall replication then passes low-level quality checks on all five validation videos but fails physical recovery with 111.7% median acceleration relative error. That validation split is retained as a failure, and the designated final split remains unopened.

The paper contributes four separable pieces of the overall argument:

1. **A rewrite-level verification contract.** Reality Probe defines how proposal evidence, reserved physical evidence, uncertainty, and a three-way SUPPORT/VETO/UNRESOLVED

decision are recorded before a captured-world rewrite is committed.

2. **A falsification-based information study.** Frozen synthetic stages show that a mathematically valid mechanism-sensitive probe can still be information-poor, and that a known-force channel can increase the evidence scale by 11.46 $\times$ without reaching the same fixed decision boundary.
3. **A locked measured-video success case.** On 10 fresh IRIS pendulum videos, the finite-amplitude probe achieves 90/110 correct nested controlled decisions, compared with 40/110 for the matched small-angle diagnostic baseline, while preserving all 10 placebo abstentions.
4. **A preserved cross-domain failure.** The frozen IRIS free-fall validation passes tracking quality on 5/5 videos but fails the physical recovery gate. Post-validation rescue analyses

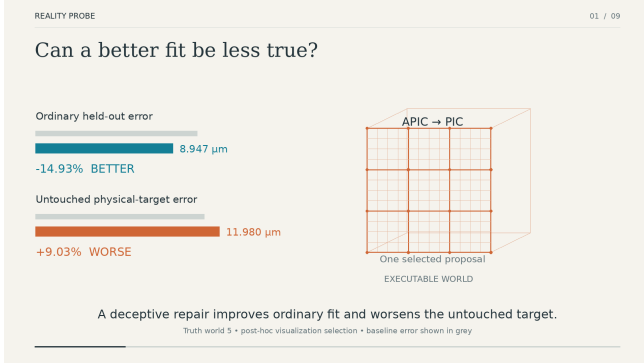


Fig. 2. Selected deceptive repair from the frozen fresh synthetic force-compliance population. Ordinary held-out error improves by 14.93%, while untouched physical-target error worsens by 9.03%. The selected case is illustrative only and does not affect aggregate statistics or decision thresholds.

do not produce a robust target-free estimator, so the lane is retired without opening its final split.

The central question throughout the paper is therefore concrete: *what physical evidence is sufficient to approve this rewrite?* The experiments show that the answer depends on the information carried by the verification channel, and that an unresolved result can be the correct endpoint of a well-functioning verifier.

2 RELATED WORK

2.1 Physics-aware scene representations and inverse physics

PAC-NeRF couples continuum simulation to a neural radiance field so that geometry and physical parameters can be estimated jointly from multi-view video [Li et al. 2023]. PhysGaussian brings Material Point Method (MPM) dynamics into 3D Gaussian primitives [Xie et al. 2024]. Subsequent systems extend physics-aware visual representations toward material initialization and refinement, interaction synthesis, and monocular or multi-material inverse physics [Liu et al. 2025; Rho et al. 2026; Wada et al. 2026; Zhang et al. 2024]. These methods establish the proposal side of our setting: a captured representation can support simulation and physical parameter inference. Reality Probe begins after such a proposal exists and asks how a particular rewrite should be verified before commit.

Differentiable simulators such as ChainQueen and DiffTaichi provide infrastructure for gradient-based physical optimization [Hu et al. 2020, 2019]. Our implementation similarly uses finite differences, adjoint sensitivity, APIC/MPM replay, and material calibration. These components support the experiments but are not presented as new simulation algorithms.

2.2 Identifiability and experiment design

Physical inference is limited by what the observations can distinguish. Structural identifiability formalizes whether model parameters are recoverable from a given observation process, and recent work studies such conditions directly from video for classes of dynamical systems [Wang et al. 2026b]. IRIS provides controlled real

videos for evaluating parameter accuracy, identifiability, extrapolation, and governing-equation recovery [Khanbayov et al. 2026]. In mechanics, optimal experiment design uses sensitivity, Jacobian conditioning, and Fisher-information ideas to choose loading conditions that separate candidate models [Asadi and Laksari 2025].

Reality Probe does not introduce a new identifiability theory or experiment-design criterion. It uses the same information question at the level of a proposed rewrite: after the candidate is fixed, does the reserved experiment separate it from the baseline by enough margin to justify a decision? The frozen unresolved outcomes in our experiments are therefore evidence about the available channel, not merely failed optimization runs.

Abstention also has a long history. Classical reject-option analysis characterizes the trade-off between recognition error and rejection [Chow 1970], while modern selective prediction learns or calibrates rejection mechanisms to control predictive risk and coverage [Geifman and El-Yaniv 2019]. Reality Probe does not claim abstention as a new decision-theory primitive. UNRESOLVED is the rewrite transaction’s explicit record that a reserved physical experiment failed to separate the baseline and candidate by the declared evidence margin.

2.3 Simulation credibility and measured physical fidelity

Verification, validation, uncertainty quantification, and intended-use credibility are established parts of computational modeling [Oberkampff et al. 2007]. Digital-twin research also considers how simulated counterfactuals should be validated [Laudy 2026]. Reality Probe adopts that credibility discipline for an individual physical rewrite in a persistent captured world. Its scope is narrower than general model certification: the system records whether a specific candidate is supported, vetoed, or left unresolved by a reserved physical channel.

GAUGE provides measurement-grounded tests of physical fidelity across simulation engines and video world models for rigid, cable, textile, and volumetric-deformation tasks [Wang et al. 2026a]. We use two GAUGE evidence classes and keep them separate. The foam-shearing study is retrospective: ordinary and marker channels prefer one model, while a longitudinal mechanism channel prefers another in 9/10 held-out repeats. A later repository-frozen validation lane uses previously untouched stretching and compression repeats 5–7. Across 12 measured worlds, the temporal annihilating witness makes 0/12 correct SUPPORT decisions and 0/12 correct VETO decisions, while all 12 placebo cases remain UNRESOLVED. Saved-output forensics show correct raw truth ordering but a signal only about 0.28% of repeat variability and 11.8% of nominal-versus-refined numerical discrepancy. We therefore treat that lane as an information-limit result and leave repeats 8–10 unopened.

2.4 Measured deformable tracking

Sequence C2 from the DOT deformable tracking dataset provides measured 3D correspondences through invisible markers [Li et al. 2025]. It is useful for exercising the captured-world reconstruction and fitting pipeline on measured deformable motion. DOT does not provide the stress-free state, applied loads, thickness, mass, or constitutive ground truth needed to identify a unique cloth material.

We therefore report fitted parameters from this check as model-conditioned effective quantities and keep them outside the main physical-verification claim.

3 PROBLEM FORMULATION

Let M denote an executable captured world. A baseline physical realization R_0 has already been fitted, and an optimizer proposes a candidate realization R_1 . The experiments assign observations to three roles. *Fitting evidence* may change the model or generate the proposal. *Ordinary held-out evidence* measures generalization within the same observation family without fitting on those samples. *Reserved physical evidence* is withheld until the candidate has been fixed and is used only for verification.

Let $L_{\text{obs}}(R)$ be the scalar ordinary held-out loss for realization R , with lower values indicating better agreement. Let $L_{\text{target}}(R)$ be the error on an untouched physical target, again with lower values preferred. We call R_1 a deceptive rewrite when

$$L_{\text{obs}}(R_1) < L_{\text{obs}}(R_0) \quad \text{and} \quad L_{\text{target}}(R_1) > L_{\text{target}}(R_0).$$

The definition identifies an evidence mismatch. It does not prescribe its cause. Model-form error, compensating parameters, boundary assumptions, transfer choices, or numerical effects can all produce a candidate that fits one observation family better and behaves worse under another physical test.

For verification, let Y denote the reserved physical evidence and $e(R, Y)$ the error between realization R and that evidence. The error may be measured in different physical units in different experiments; the uncertainty scale σ uses the same units as the numerator. We define the signed standardized progress score

$$z = \frac{e(R_0, Y) - e(R_1, Y)}{\sigma}.$$

A positive score means that the candidate is closer to the reserved evidence than the baseline. A negative score means that the candidate is worse. Dividing by σ expresses that improvement or degradation relative to the uncertainty carried by the experiment. The score is an operational standardized evidence measure; we do not interpret it as a Gaussian statistical z-test.

The frozen decision rule is

$$\begin{aligned} z \geq 2 &\Rightarrow \text{SUPPORT}, \\ z \leq -2 &\Rightarrow \text{VETO}, \\ |z| < 2 &\Rightarrow \text{UNRESOLVED}. \end{aligned}$$

The magnitude 2 is a repository-frozen decision reference for these experiments. It is neither a physical constant nor a universal credibility threshold. Keeping it fixed after data access is essential to the interpretation of the negative results: weak evidence remains weak rather than being converted into a decision by post-hoc threshold changes.

4 REALITY PROBE

4.1 Captured-world rewrite transaction

Reality Probe keeps proposal generation and verification sequential. A captured-world bundle stores appearance-linked state, particle

state, observations, uncertainty sidecars, and provenance. The fitting split calibrates the baseline and may generate a candidate rewrite. Ordinary held-out observations measure whether that candidate generalizes within the proposal evidence family. Once the candidate is fixed, the verifier evaluates a separately reserved physical channel and records the resulting evidence score and decision.

The evidence layer has three outcomes. SUPPORT means the candidate clears the positive decision margin; VETO means it clears the negative margin; UNRESOLVED means the measured separation is too small relative to the uncertainty scale. The storage transaction is binary even though the scientific decision is ternary. SUPPORT may permit a verified commit. VETO and UNRESOLVED both leave the candidate uncommitted, but the evidence record preserves the difference between contradiction and insufficient information.

This ordering protects the role of the reserved evidence. If the verification channel is consulted while the candidate is still being selected, it becomes part of the proposal objective and no longer tests the proposed rewrite on held-out physical information.

The research implementation is written in C++20 and includes nonlinear MPM replay, APIC/PIC transfer variants, finite-difference and adjoint influence paths, deterministic experiment drivers, frozen JSON/CSV result ledgers, and SHA-256-indexed evidence packages. Metal and Vulkan render paths provide system integration and presentation. The quantitative claims reported here come from the recorded experiment outputs rather than from renderer-specific behavior.

4.2 Dark-Field Counterfactual Spectroscopy

For a physical model M and intervention vector u , let $F_M(u)$ denote the observable response. Depending on the experiment, that response may contain displacement, motion, or another measured physical quantity. DCS evaluates several interventions u_i and combines their responses with signed weights w_i :

$$\mathcal{D}_\mu[F_M] = \sum_i w_i F_M(u_i).$$

The signed measure $\mu = \{(u_i, w_i)\}$ is chosen so that lower-order response terms cancel. For a declared cancellation order k , the weights satisfy

$$r_\alpha = \sum_i w_i u_i^\alpha \approx 0, \quad |\alpha| < k.$$

Here α is a multi-index that identifies a monomial in the intervention coordinates, and $|\alpha|$ is its total degree. Small r_α means that the stencil nearly annihilates that lower-order term. The implementation enumerates the required monomials and checks the largest residual against a tolerance, so the cancellation property is verified for each synthesized stencil.

A one-dimensional order-2 example is

$$F(+a) + F(-a) - 2F(0), \tag{1}$$

where a is the intervention amplitude. The constant terms cancel because the weights sum to zero, and a response that is linear in a also cancels between $+a$ and $-a$. The remaining value therefore emphasizes curvature and other higher-order response. Equation 1

is only an intuition-building example; the reported order-2 experiments synthesize their comparator on a 3×3 intervention lattice.

For finite-amplitude interactions, the implementation also evaluates the Boolean-lattice Möbius contrast

$$\kappa(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} F(T),$$

where S is a set of simultaneously active intervention factors and T enumerates its subsets. The alternating sum removes contributions already explained by lower-order subsets and leaves the finite-amplitude interaction associated with the full set. We use $\kappa(S)$ as an interaction statistic and make no claim that this classical construction is a new mixed-derivative result.

4.3 Uncertainty and standardized separation

The verifier keeps three uncertainty sources explicit. Measurement uncertainty describes error in the observed physical quantity. Repeat uncertainty measures variation across repeated acquisitions or trials. Numerical uncertainty measures sensitivity to solver resolution or discretization. For a signed DCS stencil, independent measurement and repeat variances propagate with the squared weight gain $\sum_i w_i^2$. Numerical errors can be correlated across intervention points, so D3 also measures numerical change directly in witness space by evaluating the same signed witness at nominal and refined solver resolution.

Candidate disagreement is evaluated as a standardized separation: a physical error difference divided by the combined uncertainty scale for that comparison. This normalization answers a practical question that raw error cannot: is the observed improvement large relative to the uncertainty of the experiment?

Active stencil synthesis searches within the nullspace of the lower-moment constraints. Candidate disagreement is projected into that nullspace, and the implementation evaluates a deterministic set of projected directions before retaining the strongest admissible witness. Because the resulting maximin search is non-convex, we treat this synthesis as a deterministic heuristic. The experiments evaluate whether the witnesses it produces are useful; they do not assume global optimality.

4.4 Known-force compliance as a different information channel

D4V leaves every repair unresolved under the tested kinematic verification channels. We therefore freeze a new experiment that changes the physical information supplied to the verifier. Proposal generation remains unchanged. After each candidate is fixed, a known external force is applied and the resulting compliance is measured.

The fresh population contains six APIC truth worlds. Young's modulus takes values in $\{14125, 16625, 18125\}$ Pa, Poisson ratio in $\{0.22, 0.32\}$, and the truth timestep is 2.5×10^{-5} s. Candidate families are fixed before execution. The bottom particle layer is kinematically fixed and gravity is disabled. Every top-layer particle receives a 40 N load in four separate directions: $+x$, $-x$, $+y$, and $+z$. For each load we record mean top-layer and interior-layer displacement along three axes, giving a six-component response bundle per force direction.

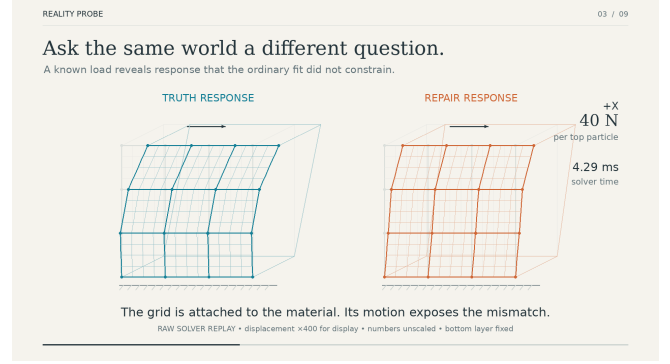


Fig. 3. The same controlled $+x$ force probe applied to the truth and repaired worlds. Displayed deformation is magnified 400 $\times$ for legibility; every reported quantitative result uses the unscaled displacement.

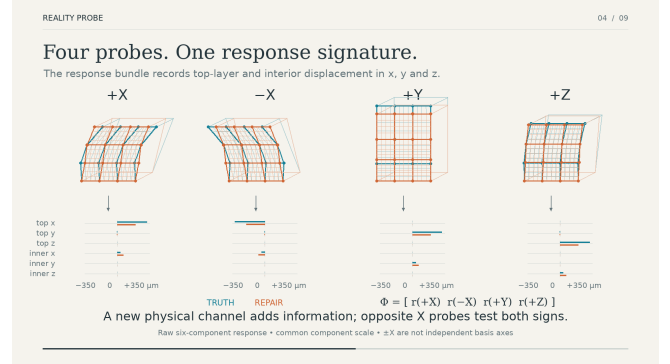


Fig. 4. Four-direction force-response fingerprint for the selected frozen case. The six-component response bundle records top-layer and interior displacement under $+x$, $-x$, $+y$, and $+z$ probes. Truth and candidate responses share one component scale.

Numerical uncertainty is the root-mean-square difference between nominal and half-timestep candidate response bundles. Measurement and repeat scales are each frozen at 2×10^{-5} m per component. If B denotes the baseline, R the candidate, and Y the truth response, the force-compliance score is

$$z_{\text{force}} = \frac{e(B, Y) - e(R, Y)}{\sqrt{\sigma_{\text{meas}}^2 + \sigma_{\text{repeat}}^2 + \sigma_{\text{num}, B}^2 + \sigma_{\text{num}, R}^2}}.$$

The numerator is positive when the candidate matches the truth response better than the baseline. The denominator combines the frozen measurement and repeat scales with baseline and candidate numerical uncertainty, all in compatible displacement-error units. The force channel and the fresh DCS comparator use the same decision magnitude of 2. Force amplitude, directions, truth worlds, candidate grid, and threshold are fixed before execution.

5 EXPERIMENTAL PROGRAM

The experiments are organized by evidence role because the same numerical outcome means different things depending on when the

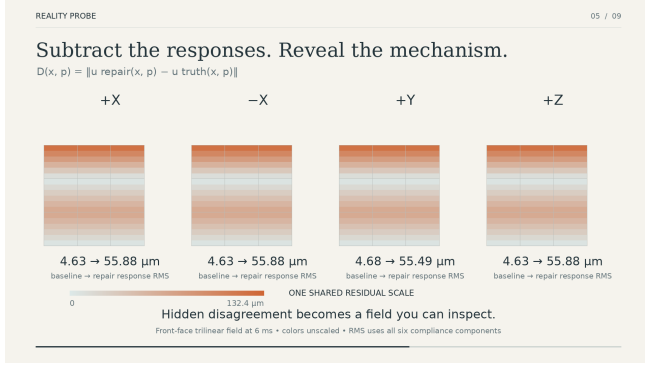


Fig. 5. Solver-driven residual field for the same frozen visualization case. All four force directions share one residual scale. The field explains where the responses differ; it is not itself the DCS decision statistic.

data were seen. Constructed controls test whether an implementation can express the intended behavior. Frozen validation tests a protocol after its choices have been fixed. Retrospective measured analyses diagnose already-observed behavior. Locked final tests use media reserved until the protocol has passed its earlier gates. Post-final analyses may probe robustness or alternative explanations, but they cannot replace the locked result.

Table 1 summarizes this evidence ladder. Here, resolved refers to the frozen SUPPORT/VETO rule. An unresolved experiment can still reveal how much information is missing or which source of uncertainty dominates.

5.1 Evidence surfaces and benchmark roles

The quantitative evidence comes from frozen JSON/CSV ledgers and measured dataset adapters. Figures expose the physical consequences of those experiments but do not create a second source of quantitative truth. Table 2 makes the role of each source explicit.

The deceptive-repair, same-probe, force-fingerprint, residual-field, signal-gain, and information-limit panels are generated from committed solver trajectories or frozen result ledgers. Display magnification is disclosed where used; numerical values are computed from the unscaled simulation outputs.

5.2 Synthetic falsification ladder

D1: constructed control. D1 deliberately builds a strong deceptive-repair signal. Its purpose is implementation sanity: if the DCS path cannot reject these 64 constructed cases, later failures are uninterpretable.

D2: frozen validation. Discovery compares raw maximin, Fisher/Jacobian selection, a same-cost raw bundle, maximum motion, DCS order-2 maximin, and random selection. After the discovery choice is fixed, D2 evaluates 16 unseen synthetic truth worlds under the frozen decision magnitude.

D3: witness-space refinement. D3 uses six fresh truth worlds to test a specific alternative explanation for weak D2 evidence: solver discretization or cancellation order may be hiding an otherwise useful distinction. The same signed witness is evaluated at nominal



Fig. 6. Held-out DOT C2 measured-versus-predicted geometry at $t = 0.167$ s in the camera-frame x/y projection. Circles are measured correspondences, crosses are predictions, and connecting segments are true-scale residuals. Across the 90 held-out samples from both nonzero checkpoints, the 3-D RMS is 4.417 mm.

and refined resolution, and adaptive selection may choose order 2 or 3.

D4V: pair-specific rewrite verification. D4V begins with 36 proposals that have already improved ordinary held-out error. The untouched physical target labels 14 as deceptive and 22 as beneficial. DCS is compared with raw-bundle, pair-specific raw, Fisher, and maximum-motion verification under the inherited $|z| \geq 2$ rule.

5.3 Measured engineering and GAUGE protocols

The DOT C2 check uses measured 3D correspondences to exercise the full captured-world path. The importer keeps measured trajectories separate from explicit proxies for rest state, density, thickness, mass, Gaussian photometry, and uncertainty. Calibration uses only the fit split, followed by held-out replay and a local material rewrite. Because C2 does not provide constitutive ground truth, the fitted material parameters are interpreted only as effective parameters of the chosen model.

GAUGE contributes two distinct measured protocols. GAUGE is retrospective in the foam-shearing analysis, which compares

Table 1. Experimental stages and evidence classes. Detailed numerical outcomes are reported in Sec. 6.

Stage	Evidence class	Population	Purpose
D1	constructed synthetic control	64 repairs	Check that the DCS path can reject a deliberately strong deceptive repair.
D2	frozen synthetic validation	16 truth worlds	Test whether the selected witness reaches the fixed decision margin on unseen truth worlds.
D3	fresh synthetic refinement	6 truth worlds	Separate numerical-resolution limits from physical observability limits.
D4V	frozen synthetic repair population	36 repairs	Evaluate pair-specific verification on ordinary-improving proposals.
DOT C2	measured engineering check	225 correspondence points	Exercise capture, fitting, held-out replay, rewrite, verification, rollback, and visualization on measured deformable motion.
GAUGE shearing	retrospective measured analysis	10 held-out repeats	Test whether ordinary and mechanism-sensitive channels can disagree on the same measured motion.
GAUGE validation	frozen measured validation	12 measured worlds / 36 cases	Test directional SUPPORT/VETO coverage on previously untouched stretching/compression repeats.
OFC	fresh synthetic orthogonal channel	36 repairs	Ask whether a different physical interrogation changes the evidence scale under the same decision magnitude.
IRIS pendulum	repository-locked measured final	10 videos / 110 nested cases	Evaluate a matched physical observable on fresh final media.
IRIS free-fall	frozen measured validation	5 validation videos	Test transfer to a second equation family without retuning on validation.

Table 2. Evidence coverage. Public measured benchmarks are separated from controlled synthetic partitions and visualization-only geometry.

Source	Evidence class	What it contributes	Claim boundary
DOT C2 [Li et al. 2025]	public measured trajectories	End-to-end captured-world ingestion, fit/held-out replay, local rewrite, verification, rollback, and measured-motion visualization.	Stress-free state, loads, thickness, mass, and constitutive ground truth are absent; fitted material values remain model-conditioned.
GAUGE [Wang et al. 2026a]	public measured benchmark	Retrospective shearing contradiction and a separately frozen stretching/compression validation lane.	The shearing study is retrospective; the validation lane is a failed validation-stage result, not final confirmation.
IRIS [Khanbayov et al. 2026]	public measured benchmark	Locked pendulum confirmation and a separately frozen free-fall validation attempt.	The 110 pendulum cases are nested in 10 videos; the free-fall lane fails validation and its final split remains unopened.
D2/D3/D4V + OFC	controlled synthetic	Known-ground-truth falsification, pair-specific rewrite evaluation, and a frozen change of physical evidence channel.	Controlled evidence, not measured-world force-sensor validation.
Stanford Bunny visual suite	visualization carrier	Solver-driven deformation, force direction, mechanism fingerprints, and residual fields.	Visualization only; it is not benchmark geometry and contributes no benchmark result.

ordinary face/marker error with a longitudinal mechanism channel over ten held-out repeats. The prospective lane is frozen separately on foam stretching and compression, soft and hard materials,

and repeats 5–7. That validation partition contains 12 measured worlds and three predeclared cases per world: SUPPORT, VETO,

Table 3. Ordinary-improving synthetic proposal populations. D4V and the fresh OFC population are separate partitions and are not pooled.

Population	Deceptive	Beneficial	Resolved at $ z \geq 2$
D4V	14	22	0/36
Fresh OFC	12	24	0/36

Table 4. DOT C2 measured-source engineering check. The material parameters are model-conditioned effective values, not material ground truth.

Quantity	Result
Measured correspondence points	225
Trajectory observations (fit / held-out)	675 (585 / 90)
Effective E, ν	7.5 kPa, 0.45
Dynamic RMS (fit / held-out)	4.391 / 4.417 mm
Adaptive candidate set	182/225 particles; 94.80% abs. gradient
Rewrite verification	nonlinear pass; oracle fail; rollback

and placebo/UNRESOLVED. Repeats 8–10 are reserved as final media and remain unopened after the validation gate fails.

5.4 Fresh orthogonal force-compliance protocol

The force-compliance experiment changes the verification channel while keeping proposal generation fixed. Fresh prospective force evidence is synthetic in this study; no measured force-sensor verification claim is made. It evaluates 36 fresh synthetic proposals, of which 12 are deceptive and 24 beneficial on the untouched target. Fresh DCS and force compliance share the same decision magnitude $\tau = 2$.

The advancement gate is frozen before execution. It requires at least four deceptive repairs, at least 30% resolved force coverage, at least 50% veto recall on deceptive repairs, at most 25% false-veto rate on beneficial repairs, force coverage greater than DCS coverage, a higher force median $|z|$, and finite numerical comparisons for every proposal. A later threshold sweep is post hoc and cannot redefine this gate.

5.5 Locked IRIS pendulum protocol and post-final analyses

The measured positive test uses IRIS pendulum_90. The tracker, candidate schedule, decision magnitude, and final media are repository-frozen before the 10-video final set is opened. The primary verifier uses the finite-amplitude pendulum period relation. The matched diagnostic baseline uses the small-angle approximation. Each physical video generates 11 controlled SUPPORT/VETO/UNRESOLVED cases, for 110 nested case rows across 10 videos.

Post-final analyses form a separate evidence class. They resample at the physical-video level, compare two additional pendulum residual models, generate GT-hidden proposals from early video segments, vary proposal/probe overlap, propagate recorded rope-length uncertainty, and stress the locked decision under image corruption

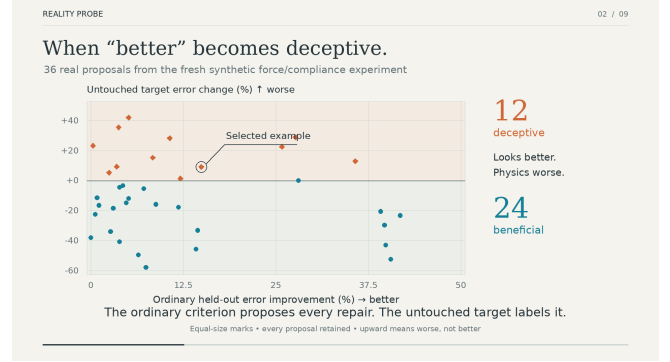


Fig. 7. Fresh force-compliance proposal population. Ordinary held-out improvement and untouched-target behavior disagree for part of the population, producing both deceptive and beneficial rewrites.

and temporal downsampling. None of these analyses can replace or retune the locked final result.

5.6 Frozen IRIS free-fall transfer test

The free-fall lane tests a different equation family with a separately frozen protocol. Development uses drop_50/06. .10; validation uses untouched drop_100/06. .10; the designated final set is drop_150/02. .10. The V6.6 policy forbids threshold retuning, tracker changes, candidate changes, or scene substitution after lock. Validation is opened only after the development gate passes. A failed validation permanently removes that split from confirmatory use and prevents the final set from being opened.

6 RESULTS AND ANALYSIS

6.1 Ordinary improvement can hide physical degradation

The deceptive-repair condition appears in both controlled ordinary-improving populations. D4V contains 14 deceptive repairs among 36 proposals; the fresh force-compliance population contains 12 among 36. These fractions describe the constructed candidate populations and should not be interpreted as a universal deception rate. Their role is narrower: an approval rule based only on ordinary held-out improvement would accept candidates that are worse on an untouched physical target.

Figure 2 shows one such case. The candidate lowers ordinary held-out error by 14.93% while increasing untouched-target error by 9.03%. The measured observation is the disagreement between those two evaluation channels. The interpretation is that ordinary held-out fit alone is insufficient for deciding whether this physical rewrite should be committed.

6.2 DCS succeeds in the control and remains information-poor prospectively

D1 confirms that the implementation can express a strong dark-field rejection: all 64 constructed deceptive repairs improve the ordinary criterion and all 64 are rejected by DCS. Median ordinary improvement is 60.88%, and the median witness-degradation ratio is 9.375.

Table 5. Retrospective GAUGE shearing comparison. Lower error is better. The sign test is post hoc.

Metric	Endpoint	Overlap
Median marker error (m)	0.002428	0.001709
Median longitudinal error	0.003131	0.005584
Longitudinal held-out wins	9/10	1/10

The prospective stages are much weaker. In D2 discovery, ranking agreement is 95.83% for raw maximin, 87.50% for Fisher/Jacobian selection, 75.00% for the same-cost raw bundle, 66.67% for maximum motion, 62.50% for DCS order-2 maximin, and 45.83% for random selection. DCS is therefore not the strongest matched ranking method in that comparison. On the 16 frozen D2 truth worlds, no witness reaches the decision magnitude; median standardized separation is 0.041578. Median numerical RMS is 6.469×10^{-6} m and the maximum moment residual is 3.886×10^{-16} . The cancellation algebra is numerically well behaved while the physical alternatives remain weakly separated.

D3 tests whether solver refinement recovers the missing separation. Adaptive selection chooses order 2 on all six fresh worlds, and the median standardized separation is 0.0588583. Refinement lowers median numerical RMS from 8.435×10^{-7} m for order 2 to 2.153×10^{-7} m for order 3, yet no world is resolved. Ranking agreement remains modest: Fisher 61.11%, raw bundle 58.33%, maximum motion 58.33%, pair-aware raw 55.56%, adaptive DCS 52.78%, and order-3 DCS 47.22%.

This separates numerical accuracy from physical observability. The refined witness is numerically cleaner, but the evidence available to distinguish the candidate worlds barely changes.

D4V reaches the same decision-level outcome on ordinary-improving proposals. DCS, raw bundle, pair-specific raw, Fisher, and maximum-motion verification all have 0/36 resolved coverage at $|z| \geq 2$. The post-hoc information-frontier calculation quantifies the remaining gap: median DCS magnitude is 0.05024, corresponding to roughly 39.8 $\times$ median amplification to reach magnitude 2 if the uncertainty scale were unchanged. This number is a diagnostic of missing evidence, not a revised threshold.

6.3 Measured GAUGE channels disagree and remain below the frozen decision scale

The retrospective GAUGE shearing analysis shows a measured channel contradiction. Ordinary face-plus-marker metrics prefer the finite-overlap candidate in all ten repeats, and marker dark-field agrees. The longitudinal mechanism channel prefers the endpoint model in 9/10 repeats. Median marker error is 0.002428 m for endpoint and 0.001709 m for overlap, whereas median longitudinal error is 0.003131 for endpoint and 0.005584 for overlap. A post-hoc exact two-sided sign test on the longitudinal direction gives $p = 0.02148$.

The separately frozen GAUGE validation lane provides a prospective measured failure. Across 12 stretching/compression worlds, the temporal annihilating witness makes 0/12 correct SUPPORT decisions and 0/12 correct VETO decisions, while all 12 placebo cases remain UNRESOLVED. Saved-output forensics show correct raw truth ordering on every directional case. The median raw signal

Table 6. Fresh synthetic force-compliance results under the common frozen decision magnitude.

Quantity	DCS	Force compliance
Median $ z $	0.04877	0.55863
Maximum $ z $	0.17279	1.31911
Resolved coverage	0/36	0/36
Median factor to $ z = 2$	41.01 $\times$	3.58 $\times$
Best-case factor to $ z = 2$	11.57 $\times$	1.52 $\times$

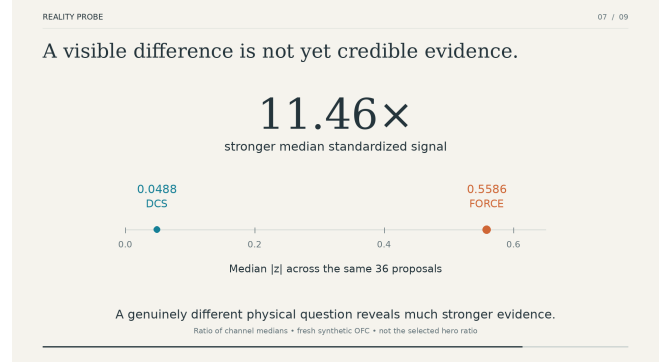


Fig. 8. Fresh known-force compliance raises median standardized evidence magnitude by 11.46 $\times$ relative to fresh DCS under the same decision rule.

is only 0.00277 times the repeat variability and 0.1179 times the nominal-versus-refined numerical discrepancy. The evidence therefore points to an information-limited channel under the frozen probe rather than a reversal of the underlying truth ordering. Repeats 8–10 remain unopened.

6.4 Known-force compliance changes the evidence scale but does not clear the gate

Fresh DCS on the orthogonal-information population has median $|z| = 0.04877$ and maximum $|z| = 0.17279$. Known-force compliance raises the median to 0.55863 and the maximum to 1.31911. The aggregate median-magnitude ratio is 11.455 $\times$.

The evidence scale changes substantially, but the frozen advancement gate still fails. Force compliance resolves 0/36 proposals, so resolved coverage and deceptive-repair veto recall do not meet their required minima. The population-size, beneficial false-veto, median-signal, and numerical-comparison conditions pass. The observation is therefore precise: this physical channel carries more discriminating information than fresh DCS, but the measured separation remains insufficient for a SUPPORT/VETO decision at $\tau = 2$.

A post-hoc threshold sweep checks how quickly that conclusion changes as the decision magnitude is relaxed. Fresh DCS has no crossing even at $\tau = 1$. Force compliance resolves 14/36 proposals at $\tau = 1$: six SUPPORT and eight VETO. Five of the 12 deceptive repairs are vetoed, giving 41.7% recall, and three of the 24 beneficial repairs are falsely vetoed, giving 12.5% false-veto rate. At $\tau = 1.5$ and above, coverage returns to 0/36. The sweep describes the score distribution; it does not redefine the prospectively frozen policy.

Table 7. Post-hoc threshold sensitivity from the frozen per-proposal ledger. The $\tau = 2$ row is the repository-frozen primary rule.

Decision magnitude τ	DCS crossings	Force crossings
1.0	0/36	14/36
1.5	0/36	0/36
2.0	0/36	0/36
2.5	0/36	0/36
3.0	0/36	0/36

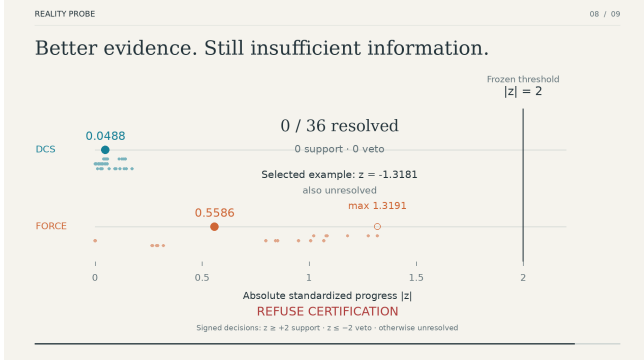


Fig. 9. Information-frontier view of the fresh synthetic experiment. Force compliance moves the evidence closer to the frozen decision magnitude, but even the strongest proposal remains below it.

6.5 The locked pendulum test provides a measured positive case

All ten final pendulum_90 videos pass the quality gate. Across the 110 controlled cases nested within those videos, the finite-amplitude probe makes 90 decisions correctly (81.82%). The small-angle diagnostic baseline makes 40/110 correctly (36.36%). The primary probe is correct on 40/50 SUPPORT cases, 40/50 VETO cases, and all 10 UNRESOLVED placebo cases. In the paired case table, the primary probe is correct when the baseline is not on 50 cases; the reverse occurs on zero cases, with 60 ties. The locked physical-recovery diagnostic has 16.81% median relative error in period-inferred pendulum length.

The physical video is the correct inference unit. A post-final clustered analysis therefore resamples the ten videos rather than the 110 nested rows. Per-video strict accuracies are identical across the ten final videos, giving primary accuracy 0.8182, baseline accuracy 0.3636, and a difference of 0.4545 in every video-level bootstrap resample. More directly, the primary probe beats the baseline on 10/10 videos with no ties; the exact two-sided sign-test value is $p = 0.001953125$.

Two additional post-final residual comparators do not explain away the locked result. A direct finite-amplitude period residual reaches 45.45% strict three-way accuracy, and a damped nonlinear ODE residual reaches 9.09%; both retain 100% placebo accuracy. Relative to the direct residual, the locked primary is uniquely correct on 40 cases and the comparator on zero. Relative to the damped ODE residual, the corresponding counts are 80 and zero. These

are secondary comparisons and do not replace the locked primary-baseline result.

The GT-hidden post-final proposal test removes the manually scheduled candidate factors. For each video, an early segment generates the proposal and a later segment evaluates ordinary improvement. The blind proposal file is hashed before ground truth is joined. All ten proposals improve the ordinary holdout; eight are physically worse and two are physically better. The reserved final-window verifier classifies all ten correctly. Varying proposal/probe overlap from $\rho = 0$ to $\rho = 1$ leaves these ten post-final decisions correct throughout, with median absolute score decreasing from 18.08 to 16.21. This tests dependence on temporal partition choice; it does not make two windows from the same video independent sensor modalities.

Recorded rope-length uncertainty also leaves the controlled truth labels unchanged: all 110 labels remain stable at both $\pm 1\sigma$ and $\pm 2\sigma$. The full corruption suite runs 180 condition-video combinations with no failed videos. The clean Metal path reproduces the locked decision vector on all ten videos. Tested noise, blur, frame duplication/drop, central occlusion, and crop levels preserve 81.82% strict accuracy. Temporal downsampling exposes the clear boundary: accuracy falls to 63.64% at 30 fps and 45.45% at 15 fps, while placebo accuracy remains 100%.

The measured observation is therefore specific but useful. A pendulum-specific observable supports accurate SUPPORT/VETO decisions on fresh video under the locked protocol, while the post-final stress tests identify temporal resolution as an acquisition constraint.

6.6 The free-fall transfer test fails validation

V6.6 passes its drop_50 development gate before validation is opened: 4/5 videos pass quality, there are zero implementation errors, and median acceleration relative error is 8.05%. Take 07 is excluded because no causal release-to-terminal-impact event survives the frozen selector.

The untouched drop_100/06 . . 10 validation split then fails. All five videos pass the low-level quality checks, but median acceleration relative error is 111.7%. Truth-control accuracy is 0 and direction-sign rate is 0 under the frozen logic. The selected intervals imply approximately 15.0–19.4 frames of full-fall time, whereas a 1 m release-from-rest fall at $g = 9.80665 \text{ m/s}^2$ is about 27.1 frames at 60 fps.

That discrepancy motivates a post-validation hypothesis: the selector is often finding a locally ballistic partial interval and then assigning the full known 1 m height to it. Retrospective diagnostics support the partial-flight signature but do not produce a valid repair. Global/chunk scaling, stationary endpoint anchors, fragmented-track stitching, 16 resolution/association/ranking variants, and bidirectional hold-to-hold reconstruction all fail to produce a robust target-free estimator across the spent development and validation evidence.

The frozen protocol determines the disposition. drop_100 is permanently retrospective, the rescue analyses remain diagnostic, and drop_150/02 . . 10 stays unopened. This second domain therefore contributes a boundary result, not a second success.

6.7 The sequence isolates an information problem

Across the full program, implementation validity, evidence magnitude, and decision sufficiency separate cleanly. DCS can be numerically well behaved while far below the decision scale. A different physical channel can increase standardized evidence by more than an order of magnitude and still remain unresolved. A pendulum-specific observable can cross the decision boundary reliably on fresh measured video. A free-fall estimator that appears adequate in development can fail untouched validation.

The interpretation supported by this sequence is that verification quality depends on the information supplied by the experiment, not only on the estimator applied afterward. Reality Probe makes that dependency visible because weak evidence stays unresolved and failed validation remains failed.

7 DISCUSSION

7.1 Why the channels behave differently

The synthetic sequence rules out a simple implementation explanation for the weak DCS results. D1 shows that the code path can produce strong rejection when the contrast is deliberately present. D2 and D3 show that moment cancellation and numerical refinement can remain well behaved while standardized separation stays far below the decision magnitude. D4V extends that observation to actual ordinary-improving proposals.

Changing the physical question has a larger effect than refining the original witness. Known-force compliance increases median evidence magnitude by 11.46 $\times$ on fresh synthetic worlds. The pendulum experiment goes further: a physical observable tied directly to the governing period relation produces useful decisions on fresh measured video. The free-fall lane then shows the opposite boundary. A locally plausible event detector passes development but fails when the observed interval no longer corresponds reliably to the full physical drop.

The evidence is therefore consistent with an information-matching explanation: verification works when the measured response is sensitive to the distinction between candidate worlds and is linked to the physical quantity being judged. The experiments do not prove that information content is the only limiting factor. They show that changing the evidence channel can matter more than numerical refinement, and that development success alone does not establish that the required physical correspondence will survive a new regime.

7.2 Relation to identifiability, experiment design, and simulation credibility

Identifiability asks whether observations contain enough information to determine a model; experiment design asks which intervention or measurement makes competing models easier to distinguish [Asadi and Laksari 2025; Wang et al. 2026b]. Reality Probe operates one level closer to the rewrite transaction. It assumes that a candidate already exists and asks whether reserved physical evidence supports committing that candidate.

That view also fits established verification and validation practice [Laudy 2026; Oberkampff et al. 2007]. A model can be numerically stable and observationally accurate in one regime

while remaining unsupported for a different intended use. The SUPPORT/VETO/UNRESOLVED rule makes that distinction explicit for individual captured-world rewrites. UNRESOLVED is therefore a statement about available evidence, not a generic confidence score attached after the fact.

7.3 Limitations

The strongest measured positive result comes from one IRIS pendulum_90 regime and ten physical videos. The 110 controlled case rows are nested within those videos, so the physical sample size is ten. Post-final clustering handles that dependence explicitly, but the result still does not establish broad transfer across different pendulum lengths, cameras, motion amplitudes, or acquisition settings.

Temporal resolution is a concrete acquisition limit. The post-final corruption suite preserves 81.82% strict accuracy under the tested noise, blur, frame duplication/drop, occlusion, and crop settings, but accuracy falls to 63.64% at 30 fps and 45.45% at 15 fps. Placebo abstention remains 100%. A deployment claim therefore requires an acquisition rate high enough to preserve the period information used by the probe.

The orthogonal force-compliance experiment is synthetic. It shows that changing the physical channel can increase standardized separation on fresh truth worlds, but it does not establish measured-world force-sensor verification. The GAUGE shearing result is retrospective, and the separately frozen GAUGE validation lane has zero directional decision coverage at the primary threshold.

The free-fall result is a failed second-domain replication. Low-level tracking quality passes on all five validation videos, yet physical recovery fails badly. The post-validation diagnostics suggest a partial-flight/full-height mismatch, but that explanation remains a retrospective diagnosis rather than a validated correction. The unopened drop_150 split is deliberately preserved.

Several method-level assumptions also remain. The primary synthetic experiments use one deformable MPM regime and selected APIC/PIC families. The DCS witness search is deterministic but heuristic because the projected maximin problem is non-convex. The decision magnitude 2 is an operational repository-frozen rule, not a universal physical credibility threshold. DOT C2 fitted material values are model-conditioned because the dataset does not provide the state and loading information required for unique constitutive identification.

Proposal and verification windows in the pendulum study come from the same physical video. The post-final overlap sweep shows that the tested GT-hidden decisions are stable from disjoint to fully overlapping temporal windows, but this does not make the channels independent sensor modalities. A future experiment with separately instrumented physical sensing would provide a stronger independence claim.

Finally, the repository records exact data revisions, lock commits, hashes, result ledgers, and replay scripts. An independent Ubuntu replay has reproduced the frozen pendulum development and validation readiness stages. At the time of this manuscript revision, the exact Linux replay of the locked final pendulum_90 result remains

a reproducibility-engineering audit; the canonical locked result is the original repository-recorded execution.

7.4 What should be tested next

The next experiment should change the measured information before it changes the threshold. A useful measured follow-on would apply a calibrated external force or excitation, record that response with a sensor channel that is separate from the proposal evidence, freeze the intervention and uncertainty model before final data access, and retain the same three-way decision semantics.

Broader constitutive families and contact regimes are also needed to test whether the rewrite transaction survives beyond the current MPM setting. Modal or frequency-domain excitation is especially relevant when ordinary trajectories are weakly informative because it can expose different response structure without relying on a larger post-hoc decision margin. Any new protocol should preserve failed validation splits as failed evidence rather than recycling them into development.

8 REPRODUCIBILITY AND IMPLEMENTATION NOTES

The repository separates scientific freezes by experiment rather than treating the manuscript date as a single global lock. The core DCS scientific state is recorded at commit `a9da8c0aa868`. The locked IRIS pendulum final test uses commit `9e56f06145b8`; the failed V6.6 free-fall validation uses commit `78818c3bfd87`. The public IRIS data revision is `c253822f5543`. Full immutable identifiers are retained in the machine-readable result ledgers and provenance manifests.

Machine-readable JSON/CSV ledgers store the reported outcomes, while SHA-256 manifests bind result files and evidence packages to their recorded content. The one-command research runners build the relevant code, materialize pinned public data, execute frozen stages in order, regenerate deterministic figures/tables, and record compiler, system, and GPU provenance. The paper reads numerical claims from those ledgers rather than from rounded labels in figures.

The C++20 implementation contains nonlinear MPM replay, APIC/PIC transfer variants, finite-difference and adjoint influence paths, arbitrary-order lower-moment checks, signed-response application, finite-amplitude interaction contrasts, uncertainty propagation, active witness synthesis, witness-space numerical refinement, spatial localization, and commit/rollback bookkeeping. Experiment drivers and frozen protocol files are kept separate from the core implementation so that a failed gate can remain frozen without silently changing the underlying method.

The locked pendulum result is based on ten physical videos. The result ledger explicitly records 90/110 primary correct cases, 40/110 small-angle-baseline correct cases, 40/50 SUPPORT, 40/50 VETO, and 10/10 UNRESOLVED placebo correctness for the primary probe. Post-final hardening is stored in a separate ledger and cannot replace the final-test record.

Cross-platform replay is treated as a reproducibility audit rather than a source of new evidence. The Ubuntu/current-dependency replay reproduced the frozen development and validation stages through the original readiness gate and then completed the reserved 10-video final campaign. Its discrete final outcomes match the locked ledger exactly: 10/10 videos pass quality control; the primary and

baseline make 90/110 and 40/110 correct decisions; the primary records 40/50 correct SUPPORT, 40/50 correct VETO, and 10/10 correct UNRESOLVED placebo cases; and the paired counts are 50 primary-only wins, zero baseline-only wins, and 60 ties. The continuous median period-inferred length error changes slightly, from 0.16810865 in the locked execution to 0.16793838 on Ubuntu, an absolute difference of 0.0001703 (about 0.10% relative). We therefore treat the decision-level result as independently reproduced while retaining the small scalar drift as a cross-platform numerical reproducibility limitation. No replay outcome is used to retune the tracker, candidate schedule, threshold, or frozen final result.

Generative-AI-assisted research workflow. OpenAI ChatGPT was used for code drafting and debugging, experiment orchestration and analysis scripts, visualization/manuscript tooling, and manuscript revision. AI-generated text or code was never treated as empirical evidence. Numerical claims were checked against committed result artifacts, experimental partitions and thresholds were frozen in repository state where stated, and cited sources were checked against publication records before inclusion.

Paper visualizations are generated deterministically from committed trajectories or result ledgers. Solver-driven surfaces may use an explicitly stated display magnification so that small motion is visible; reported quantitative values remain unscaled. The Stanford Bunny is used only as a visualization carrier where identified and contributes no benchmark result.

9 CONCLUSION

Reality Probe begins from a practical failure mode: a physical rewrite can improve ordinary held-out observations and still make an untouched physical behavior worse. The experiments show that the quality of the verification decision depends strongly on the physical information supplied after the proposal is fixed.

DCS provides a useful falsification example. It succeeds on a constructed control, remains numerically well behaved, and still produces little prospective separation in the tested synthetic regimes. Known-force compliance increases median standardized evidence by 11.46× without clearing the frozen decision boundary. On fresh IRIS pendulum video, a finite-amplitude period probe produces 90/110 correct nested controlled decisions across ten videos, compared with 40/110 for the small-angle diagnostic baseline, while preserving all ten placebo abstentions. The frozen free-fall transfer test then fails validation with 111.7% median acceleration relative error and remains failed.

The evidence supports a conditional systems claim. A captured-world rewrite should be committed only when a reserved physical channel separates it from the baseline strongly enough under the declared uncertainty model. When the channel is weak, the system should remain unresolved. When a protocol fails untouched validation, the appropriate action is to retire or redesign that verification lane rather than manufacture a positive result by retuning after the fact.

ACKNOWLEDGMENTS

Generative AI disclosure. OpenAI ChatGPT was used as a research and writing assistant during development of this work, including drafting and revising manuscript text, generating and debugging code, assisting with experiment orchestration and analysis scripts, preparing tables and figures, and manuscript formatting. The author reviewed the generated material, validated reported experiments and numerical evidence against the frozen research artifacts, checked the references used in the manuscript, and takes responsibility for the final content.

REFERENCES

- Amirreza Asadi and Kaveh Laksari. 2025. Optimal Experimental Design for Repeatable Hyperelastic Material Characterization. *Journal of the Mechanical Behavior of Biomedical Materials* 170 (2025), 107104. <https://doi.org/10.1016/j.jmbbm.2025.107104>
- C. K. Chow. 1970. On Optimum Recognition Error and Reject Tradeoff. *IEEE Transactions on Information Theory* 16, 1 (1970), 41–46. <https://doi.org/10.1109/TIT.1970.1054406>
- Yonatan Geifman and Ran El-Yaniv. 2019. SelectiveNet: A Deep Neural Network with an Integrated Reject Option. In *Proceedings of the 36th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 97)*. 2151–2159.
- Yuanming Hu, Luke Anderson, Tzu-Mao Li, Qi Sun, Nathan Carr, Jonathan Ragan-Kelley, and Frédo Durand. 2020. DiffTaichi: Differentiable Programming for Physical Simulation. In *International Conference on Learning Representations*.
- Yuanming Hu, Jiancheng Liu, Andrew Spielberg, Joshua B. Tenenbaum, William T. Freeman, Jiajun Wu, Daniela Rus, and Wojciech Matusik. 2019. ChainQueen: A Real-Time Differentiable Physical Simulator for Soft Robotics. In *IEEE International Conference on Robotics and Automation (ICRA)*. 6265–6271. <https://doi.org/10.1109/ICRA.2019.8794333>
- Rasul Khanbayov, Mohamed Rayan Barhdadi, Erchin Serpedin, and Hasan Kurban. 2026. IRIS: A Real-World Benchmark for Inverse Recovery and Identification of Physical Dynamic Systems from Monocular Video. In *European Conference on Computer Vision (ECCV)*. 366–383. arXiv:2603.16432
- Olav Laudy. 2026. The Digital Twin Counterfactual Framework: A Validation Architecture for Simulated Potential Outcomes. arXiv:2604.01325
- Xinyuan Li, Yu Guo, Yubei Tu, Yu Ji, Yanchen Liu, Jinwei Ye, and Changxi Zheng. 2025. Textureless Deformable Object Tracking with Invisible Markers. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 47, 9 (2025), 7243–7254. <https://doi.org/10.1109/TPAMI.2024.3463422>
- Xuan Li, Yi-Ling Qiao, Peter Yichen Chen, Krishna Murthy Jatavallabhula, Ming C. Lin, Chenfanfu Jiang, and Chuang Gan. 2023. PAC-NeRF: Physics Augmented Continuum Neural Radiance Fields for Geometry-Agnostic System Identification. In *International Conference on Learning Representations*. arXiv:2303.05512
- Zhuoman Liu, Weicai Ye, Yan Luximon, Pengfei Wan, and Di Zhang. 2025. Unleashing the Potential of Multi-modal Foundation Models and Video Diffusion for 4D Dynamic Physical Scene Simulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 11016–11025.
- William L. Oberkamp, Martin Pilch, and Timothy G. Trucano. 2007. *Predictive Capability Maturity Model for Computational Modeling and Simulation*. Technical Report SAND2007-5948. Sandia National Laboratories. <https://doi.org/10.2172/976951>
- Daniel Rho, Jun Myeong Choi, Matthew Thornton, Biswadip Dey, and Roni Sengupta. 2026. MonoPhysics: Estimating Geometry, Appearance, and Physical Parameters from Monocular Videos. arXiv:2605.30320
- Norika Wada, Kohei Yamashita, Ryo Kawahara, and Ko Nishino. 2026. M-PhyGs: Multi-Material Object Dynamics from Video. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. arXiv:2512.16885
- Shuai Wang, Yaxin Feng, Xuekun Jiang, Shihan Tian, Ningyu Yan, Xing Shen, Chaoyang Lyu, Hui Wang, Yunsong Zhou, Hanqing Wang, Jiangmiao Pang, Yang Xiang, Xing Gao, Chunhua Shen, and Weinan Zhang. 2026a. GAUGE: A Measurement-Grounded Benchmark for Physical Fidelity in Simulation Engines and Video World Models. arXiv:2608.05948
- Yuanyuan Wang, Wenjie Wang, Kun Zhang, and Mingming Gong. 2026b. Physics from Video: Identifiability of Time-Invariant Second-Order ODEs under Minimal Trajectory Conditions. arXiv:2606.00115
- Tianyi Xie, Zeshun Zong, Yuxing Qiu, Xuan Li, Yutao Feng, Yin Yang, and Chenfanfu Jiang. 2024. PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 4389–4398.
- Tianyuan Zhang, Hong-Xing Yu, Rundi Wu, Brandon Y. Feng, Changxi Zheng, Noah Snively, Jiajun Wu, and William T. Freeman. 2024. PhysDreamer: Physics-Based Interaction with 3D Objects via Video Generation. In *European Conference on Computer Vision (ECCV)*. 388–406. https://doi.org/10.1007/978-3-031-72627-9_22